

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Biogeochemical cycles: carbon, nitrogen, phosphorus - Set 1

Mark scheme

AQA spec 3.2.4; The physical environment; textbook pp. 148-157

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	1 mark for the correct answer. • 7×10^{11} tonnes (1)	1
01.2	A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown. • 10% of 10 000 = 1000 billion tonnes (1) • $1000 / 700 \times 100 = 143\%$ (accept 142.9) (1)	2
01.3	1 mark for each point. • Carbon is transferred into and out of rocks extremely slowly (fossilisation / sedimentation, weathering, volcanic release) (1) • Residence time is millions of years so the flux to the atmosphere is tiny compared with respiration / photosynthesis / combustion (1)	2
01.4	1 mark for each point. • CO ₂ dissolves in the sea (forming carbonic acid / hydrogen carbonate ions) and is absorbed by phytoplankton through photosynthesis (1) • Organisms such as molluscs, corals and planktonic organisms build calcium carbonate shells / exoskeletons (1) • When they die the shells sink and sediment on the seabed, becoming limestone / chalk over long periods (1)	3
01.5	1 mark for each point. • Aerobic respiration uses oxygen to break down organic compounds completely to CO ₂ (and water), releasing much more energy (1) • Without oxygen, anaerobic organisms only partly break down organic compounds, releasing carbon as methane (CH ₄), which is later oxidised to CO ₂ in the atmosphere (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 148-149; AQA spec 3.2.4.2.	

Question	Marking guidance	Marks
02.1	<i>2 marks for each point (partial credit for a partly correct point).</i> • Ploughing: increases the oxygen supply to soil decomposers so aerobic decomposition of dead organic matter is faster; carbon in the soil DOM store falls and atmospheric CO₂ rises (2) • Draining: waterlogged anaerobic conditions preserved dead organic matter (peat) and produced methane slowly; drainage makes the soil aerobic so stored carbon is rapidly decomposed to CO₂ (though methane release stops) (2) • Marine pollution: kills phytoplankton so less dissolved CO₂ is absorbed by photosynthesis and less carbon sinks to sediments in dead plankton; less CO₂ then dissolves from the atmosphere to replace it, so atmospheric CO₂ rises (2)	6
02.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Rice paddy fields (1) • Landfill sites (1) • Anaerobic sediments in reservoirs (1) • Livestock intestines (cattle) (1) • Manure / slurry stores (1)	3
02.3	<i>1 mark.</i> • More CO₂ dissolves in the sea, forming carbonic acid which dissociates into hydrogen carbonate and hydrogen ions; more H⁺ lowers the pH (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 150-151; AQA spec 3.2.4.2.	
03.1	<i>1 mark for each point.</i> • Trees absorb CO₂ by photosynthesis and store carbon in wood as they grow (1) • Once mature there is no further net storage (photosynthesis balanced by respiration / decomposition), though the standing forest remains a reservoir (1) • Harvesting the timber and keeping it as wood (eg house construction) then replanting produces additional storage as the new trees grow (1)	3
03.2	<i>1 mark for each point.</i> • Pre-combustion: the fuel is modified before use so no CO₂ is released when burnt - eg gasification of coal to hydrogen and CO₂ (CO₂ stored, hydrogen distributed); oxy-fuel combustion in pure oxygen giving only CO₂ and water vapour which is condensed out (1) • Advantage: capture at a large central facility; suits small dispersed sources (vehicles) that cannot capture CO₂ (1) • Post-combustion: CO₂ removed from exhaust gases by solvents, membrane filtration, adsorption / desorption, cryogenic separation, graphene (1) • Disadvantage: only about 20% of flue gas is CO₂ so separation is costly / energy-intensive; none fully developed (1)	4
03.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Injected into depleted oil / gas fields or saline aquifers (1) • Injected into oil reservoirs to provide pressure for secondary oil recovery (1) • Reaction with fractured basalt to form solid carbonate minerals (1) • Concern: leakage back to the atmosphere / monitoring over centuries; limited suitable geology; cost (1)	3

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 151-152; AQA spec 3.2.4.2.	
04.1	<i>1 mark for each point.</i> • 1 nitrogen fixation; 2 nitrification (1) • 3 denitrification; 4 ammonification (1) • 5 ionisation; 6 leaching (3 marks for all six, 2 for four or five, 1 for two or three) (1)	3
04.2	<i>1 mark for each point.</i> • Drainage makes the soil aerobic (1) • Aerobic nitrifying bacteria (and nitrogen fixers) increase so more ammonium is converted to nitrate (1) • Anaerobic denitrifying bacteria decline so less nitrate is lost as nitrogen gas (1)	3
04.3	<i>1 mark for each point.</i> • Nitrogen gas is very unreactive so most organisms cannot use it directly (1) • Only nitrogen fixation (bacteria, lightning, Haber process) converts it to usable ammonium / nitrate; the cycle recycles a small usable amount (1)	2
04.4	<i>1 mark for each point.</i> • Rhizobium bacteria in their root nodules fix atmospheric nitrogen into ammonium / proteins (1) • When the plants die / are ploughed in, decomposition and nitrification release nitrates for the following crop (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 152-154; AQA spec 3.2.4.3.	
05.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • Human = $120 + 60 + 30 = 210$; total = 325 (1) • $210 / 325 \times 100 = 64.6\%$ (accept 65%) (1)	2
05.2	<i>1 mark for each point.</i> • Nitrogen and hydrogen react to form ammonia over an iron catalyst at high temperature and pressure; ammonia then converted to nitrates for fertiliser (1) • Uses large amounts of energy from fossil fuels (also the hydrogen source), releasing CO₂ and NO_x (1) • Biological fixation by bacteria occurs at low temperature using solar energy captured by plants (enzymes lower activation energy) (1)	3
05.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Eutrophication of water bodies by leached nitrates / sewage (1) • Photochemical smogs and tropospheric ozone from NO_x (1) • Acid rain (nitric acid) (1) • Nitrous oxide as a greenhouse gas (1) • NO_x toxicity / respiratory disease (1) • Nitrate in drinking water (methaemoglobinaemia) (1)	3
05.4	<i>1 mark for each point.</i> • Carbon is lost as CO₂ (composting) or CH₄ (digestion) so the C:N ratio improves / more nitrogen-rich (1) • Bulk and odour reduced, pathogens killed, easier to apply; digestion also yields biogas (1)	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 154-155; AQA spec 3.2.4.3.	
06.1	<i>1 mark for each point.</i> • There is no gaseous / atmospheric reservoir of phosphorus (no equivalent of nitrogen fixation) (1) • Most phosphorus compounds have low solubility so little is available in soil water (1) • Phosphorus lost to marine sediments only returns through very slow processes: mountain building (uplift) and weathering of rocks (1)	3
06.2	<i>1 mark for each point.</i> • Fungal hyphae form a symbiotic relationship with plant roots and spread through a large volume of soil (1) • Greatly increasing the surface area for absorbing (poorly mobile) phosphate ions, passed to the plant in exchange for carbohydrates (1)	2
06.3	<i>1 mark for each point.</i> • Mining of calcium phosphate rock (and past mining of guano deposits, eg Peru, Nauru, now largely exhausted) (1) • Treated to make more soluble ammonium phosphate fertilisers, mobilising phosphorus into the environment (1) • Fertiliser runoff / eroded soil / sewage effluent carry phosphates into water bodies causing eutrophication (with nitrates); harvesting removes phosphorus from farmland (1)	3
06.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Use biological wastes (manure, sewage sludge, compost) as fertiliser to return phosphates rather than mining rock (1) • Breed crops that absorb phosphate more efficiently (1) • Maintain conditions for mycorrhizal fungi (less tillage, fewer fungicides, organic matter) (1) • Recover phosphate from sewage effluent by iron(III) sulfate precipitation; reduce soil erosion (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 156-157; AQA spec 3.2.4.4.	

Question	Marking guidance	Marks
07.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) - Carbon: combustion of fossil fuels and biomass moving carbon from long-term stores to the atmosphere; deforestation reducing photosynthesis; ploughing / drainage increasing aerobic decomposition; new anaerobic environments (paddies, landfill, reservoirs, livestock) releasing methane; marine pollution reducing phytoplankton; consequences: rising CO₂ and CH₄, climate change, ocean acidification, reduced soil organic matter - Nitrogen: Haber process doubling fixation; legume cultivation; drainage increasing nitrification and reducing denitrification; combustion NO_x; sewage and fertiliser leaching; consequences: eutrophication, smogs, acid rain, N₂O warming, NO_x toxicity - Counter-strategies carbon: renewables / nuclear replacing fossil fuels, energy conservation, afforestation matched to deforestation and timber use, conserving peat bogs and forests, increasing soil organic matter (low tillage, mulching), CCS (pre / post combustion, storage) - Counter-strategies nitrogen: reduced combustion, catalytic converters / urea sprays, biological fixation instead of Haber, legumes and crop rotation, management of biological wastes as fertiliser, low-solubility fertilisers, timing, buffer strips, low tillage, nitrate vulnerable zones - Evaluation: some strategies well established and cheap (rotation, buffer strips), others expensive or unproven (CCS); scale of fossil fuel dependence; feedback and time lags; link to circular economy and low-temperature natural processes; judgement	9
07.2	<i>1 mark.</i> The average time an atom / molecule remains in the reservoir before being transferred out (volume / transfer rate) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 148-157; AQA spec 3.2.4.	
08.1	<i>1 mark for the correct answer.</i> $(120 + 95) - (2 + 45) = 168 \text{ t ha}^{-1}$ (1)	1
08.2	<i>1 mark for each point.</i> - Annual ploughing aerates the soil so aerobic decomposers break down dead organic matter faster, releasing CO₂ (1) - Harvest removes biomass so less dead organic matter is returned; pasture roots and litter constantly add organic matter (1)	2

Question	Marking guidance	Marks
08.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Low / zero tillage (1) • Adding manure, compost or mulch (1) • Growing cover crops / leaving crop residues (1) • Converting to long-term grass or tree crops in rotation (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 148-151; AQA spec 3.2.4.2.	